

Synchronization of Complex Networks via Aperiodically Intermittent Pinning Control

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Abstract—In this note, synchronization for linearly coupled network is investigated by pinning a simple aperiodically intermittent controller. Nodes' dynamical behaviors in the network are described by continuous-time ordinary differential equations. Some sufficient conditions to guarantee global synchronization are presented. Furthermore, an adaptive algorithm is designed for the pinning control gain and its validity is also proved rigorously. Finally, numerical simulations are given to demonstrate the correctness of obtained results.

Index Terms—Adaptive control, aperiodic, intermittent pinning, synchronization.

I. INTRODUCTION

In recent years, an increasing interest has been devoted to the study of complex networks from various research areas, such as communication, physical science, mathematics, sociology, and so on. Among them, a hot topic is synchronization, which can be used to explain natural phenomena appearing image processing, secure communication and so on.

In general, coupled identical systems can be described as

$$\dot{x}_j(t) = f(x_j(t)) + \eta_j(x_1(t), \dots, x_N(t)) \quad (1)$$

where $x_j(t) = (x_j^1(t), \dots, x_j^n(t))^T \in \mathbb{R}^n$ is the n -dimensional state variable of the j -th node, $j = 1, \dots, N$. $\dot{\pi}(t) = f(\pi(t))$ denotes the dynamics of each isolated (uncoupled) node, $f(\cdot) = (f_1(\cdot), \dots, f_n(\cdot))^T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuous. Continuous function $\eta_j(x_1(t), \dots, x_N(t))$ represents the diffusion coupling added on the j -th node from other nodes, which satisfies that: when $x_1 = \dots = x_N$, $\eta_j(x_1, \dots, x_N) = 0$, $j = 1, \dots, N$.

If $\|x_j(t) - x_k(t)\| \rightarrow 0$, $j, k \in 1, \dots, N$, as $t \rightarrow +\infty$, then the network (1) is said to be synchronized. In particular, when $f(\cdot) \equiv 0$, $n = 1$, it becomes the consensus problem ([1]–[3]). Many approaches have been derived for synchronization, see [4]–[9] and references therein.

Manuscript received August 7, 2013; revised April 4, 2014, October 10, 2014, and February 28, 2015; accepted March 13, 2015. Date of publication March 27, 2015; date of current version December 1, 2015. This work was supported by the National Science Foundation of China under Grants 61203149, 61273211, and 61233016, the National Basic Research Program of China (973 Program) under Grant 2010CB328101, the “Chen Guang” project supported by Shanghai Municipal Education Commission and Shanghai Education Development Foundation under Grant 11CG22, the Fundamental Research Funds for the Central Universities, and the Program for Young Excellent Talents in Tongji University. Recommended by Associate Editor G. E. Dullerud.

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Digital Object Identifier 10.1109/TAC.2015.2416912

In this note, we will investigate the synchronization of coupled systems via external controls

$$\begin{cases} \dot{x}_j(t) = f(x_j(t)) + \eta_j(x_1(t), \dots, x_N(t)) + u_j; & j \in J \\ \dot{x}_{j'}(t) = f(x_{j'}(t)) + \eta_{j'}(x_1(t), \dots, x_N(t)); & j' \notin J \end{cases}$$

where J is a subset of $\{1, 2, \dots, N\}$ denoting the controlled nodes set, u_j denotes the external control on node $j \in J$. If only a small fraction of network nodes are controlled, then it is called the pinning control, see [10]–[15]. In these papers, a variety of pinning control schemes is proposed and various criteria for node selection and the connections between controllability and network structure are given.

However, in these works, external controls are needed all the time. In practice, it may be non-economic. For example, in signal transmission, the signal will become weak due to diffusion, so an external control should be added until the strength of the signal reaches an upper level. Then, the external control can be removed. In the building energy consumption, energy saving modification of heating system will bring economic and social benefits, so that the heating system of central air-conditioning can run by the intermittent operation mode (off-preheat-occupancy-off). Another example is the generation of wind power in smart grid, whose generation is typically intermittent.

Compared with continuous control, discontinuous control strategies for synchronization have attracted great interests and been widely used in practice. It includes impulsive control, sample-data control [16], node-to-node pinning control [17]–[20], fast-switching strategy [21], edge snapping pinning control ([22], [23]), and the event-triggered scheduling algorithm ([24]–[31]). For example, in [21], the authors show that global synchronization is possible if the switching period is sufficiently small and if the oscillators globally synchronize under a time-averaged coupling. Based on performing sensing and actuation periodically, sampled-data control is proposed in the last century, which can be regarded as time-triggered control and easy to design and analyze. However, this method also leads to unnecessary waste of resources in the sense that the execution is more frequent than necessary. Therefore, in order to overcome the limitations of resource, computation and communication, event-triggered control and self-triggered control schemes are proposed recently. For event-triggered control, an event is triggered if the triggering condition is satisfied, so it is reactive and continuous monitoring is needed; on the other hand, for self-triggered control, it predicts next event time based on information at previous instant, so it is proactive and not necessary to monitor triggering condition continuously. For both cases, the triggering instants are aperiodic. In particular, [26] proposes a feedback updating law that is triggered when the error becomes large, while [30] discusses event-trigger and self-trigger approaches in both centralized and distributed formulations in the case of undirected and connected graphs.

Recently, a periodically intermittent control strategy is introduced and its applications on synchronization are also studied, see [32]–[37]. However, the requirement of periodicity is unreasonable and unnecessary in practice, and aperiodically intermittent is more general in many real cases. For example, the generation of wind power is typically aperiodically intermittent. Moreover, investigating the robustness of periodical intermittent control with some stochastic disturbance is

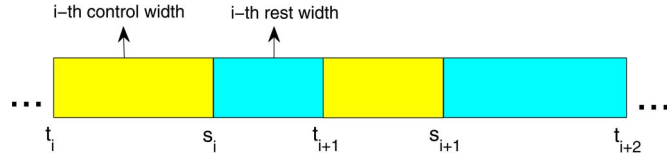


Fig. 1. Sketch map of the aperiodically intermittent control strategy.

of great significance, which also requires the study of aperiodically intermittent case. Furthermore, the intermittent control scheme fills the gap between continuous control and impulsive control, while the usual impulsive control requires aperiodic property for its control time period. Therefore, in both application and theoretical analysis, it is of significance to investigate the synchronization problem under aperiodically (or non-periodically) intermittent control.

To the best of our knowledge, there have been rare works addressing the dynamical behaviors under aperiodically intermittent control strategy [38], [39]. In this note, we will investigate the following type of aperiodically intermittent control strategy, see Fig. 1. For any time span $[t_i, t_{i+1})$, $[t_i, s_i]$ is the work time (control time), and $s_i - t_i$ is called the i -th control width (control duration); while (s_i, t_{i+1}) is the rest time, and $t_{i+1} - s_i$ is called the i -th rest width. For example, when $t_i - s_i \equiv \theta$, and $t_{i+1} - t_i \equiv \omega$, the intermittent control type becomes the periodic type, which has been investigated in [37].

In the following, for the framework of aperiodically intermittent control strategy, we will focus on the following two problems. Problem 1: Can we find sufficient conditions for networks to synchronize? Problem 2: Can we give theoretical results for adaptive aperiodically intermittent control strategy?

The rest of this note is organized as follows. In Section II, some necessary definitions, lemmas and notations are given. In Section III-A, the network model with aperiodically intermittent control for linearly coupled networks is firstly established. Some sufficient criteria for aperiodically intermittent control with static control gain are given in Section III-B. Then, we apply the adaptive approach [7] to the intermittent control gain, design a centralized adaptive rule, and rigorously prove its validity for synchronization in Section III-C. Moreover, numerical simulations to show the effect of obtained theoretical results are also presented in Section IV. Finally, the note is concluded in Section V.

II. PRELIMINARIES

Definition 1: (See [6], [9]) Matrix $A = (a_{jk})_{j,k=1}^N$ is said to belong to class $A1$, denoted as $A \in A1$, if A is irreducible, and $a_{jk} \geq 0, j \neq k, a_{jj} = -\sum_{k=1, k \neq j}^N a_{jk}, j = 1, \dots, N$.

If $A \in A1$ is symmetric, then we say that matrix A belongs to class $A2$, denoted as $A \in A2$.

Lemma 1: (See [9]) If $N \times N$ matrix $A \in A1$, then $\text{rank}(A) = N - 1$, i.e., 0 is an eigenvalue of A with multiplicity 1, and all nonzero eigenvalues of A have negative real part.

Definition 2: A network with N nodes, whose states are denoted by $x_j(t), j = 1, 2, \dots, N$, is said to synchronize to the pinning trajectory $\pi(t)$, if $\lim_{t \rightarrow +\infty} \|x_j(t) - \pi(t)\| = 0$.

Definition 3: (See [6], [9]) The function $f(\cdot)$ is said to satisfy $f(\cdot) \in \text{QUAD}(P, \Delta)$, if there exist two positive definite diagonal matrices $P = \text{diag}(p_1, \dots, p_n)$ and $\Delta = \text{diag}(\delta_1, \dots, \delta_n)$, such that for any $x, \hat{x} \in R^n$, the following condition holds: $(x - \hat{x})^T P(f(x) - f(\hat{x}) - \Delta x + \Delta \hat{x}) \leq 0$.

It can be shown that the QUAD assumption holds for several well-known chaotic oscillators, such as some cellular neural networks, the Lorenz system, and so on.

Throughout this note, we denote the identity matrix by I with appropriate dimensions. If all eigenvalues of a matrix $A \in R^{N \times N}$ are real, then we sort them as $\lambda_1(A) \leq \lambda_2(A) \leq \dots \leq \lambda_N(A)$. We denote the symmetrical part of A as $A^s = (A + A^T)/2$. A symmetric real matrix A is positive definite (semi-definite) if $x^T A x > 0 (\geq 0)$ for all nonzero x , denoted as $A > 0 (A \geq 0)$. The Kronecker product of an N by M matrix $A = (a_{ij})$ and a p by q matrix B is $A \otimes B$.

III. MAIN RESULTS

A. Network Model With Aperiodically Intermittent Control

At first, we introduce the linearly coupled network model with aperiodically intermittent control. For simplicity, we assume that the pinning control is only imposed on the first node with a constant control gain. Thus, the model now can be described as

$$\begin{cases} \dot{x}_1(t) = f(x_1(t)) + \sum_{k=1}^N a_{1k} \Gamma x_k(t) \\ \quad + c \cdot \Gamma (\pi(t) - x_1(t)); \\ \quad \text{if } t \in [t_i, s_i], i = 0, 1, 2, \dots \\ \dot{x}_j(t) = f(x_j(t)) + \sum_{k=1}^N a_{jk} \Gamma x_k(t); j = 1, \dots, N \end{cases} \quad (2)$$

Positive semi-definite matrix $\Gamma = \text{diag}(\gamma_1, \gamma_2, \dots, \gamma_n)$ denotes the inner coupling matrix, while the matrix A is the outer coupling matrix reflecting the network topology.

The target trajectory $\pi(t)$ satisfies $\dot{\pi}(t) = f(\pi(t))$.

In this model, we assume the coupling matrix $A = (a_{jk}) \in R^{N \times N} \in A1$, i.e., the corresponding network can be directed. Scalar $c > 0$ is the linear state feedback control gain.

For any node $j = 1, 2, \dots, N$, denote $e_j(t) = x_j(t) - \pi(t)$ as the synchronization error, and $E(t) = (e_1(t)^T, \dots, e_N(t)^T)^T$, $F(E(t)) = ((f(x_1(t)) - f(\pi(t)))^T, \dots, (f(x_N(t)) - f(\pi(t)))^T)^T$. Then, linearly coupled systems (2) with static intermittent pinning control can be rewritten as

$$\begin{cases} \dot{E}(t) = F(E(t)) + (\hat{A} \otimes \Gamma)E(t), & \text{if } t \in [t_i, s_i] \\ \dot{E}(t) = F(E(t)) + (A \otimes \Gamma)E(t), & \text{if } t \in (s_i, t_{i+1}) \end{cases} \quad (3)$$

where matrix $\hat{A}$ is defined as

$$\hat{A} = A - \text{diag}(c, 0, \dots, 0). \quad (4)$$

Remark 1: In fact, the coupling matrix can be rank-reduced (e.g., [9]); the control gain can also be fast switching, see [21], the feedback gain $c(t)$ can be random piecewise constant in every time interval.

Remark 2: In this note, we generalize the periodically intermittent control to aperiodic case, which is an important and necessary step in the intermittent control research. In fact, in the history of feedback control, as pointed out in [29], there exists also a wave of debate about the periodic (sampled-data) control vs aperiodic (event-triggered) control. Nevertheless, event-based control has been an important technique, which means that the aperiodic control should be discussed.

On the other hand, the aperiodically intermittent control studied in this note is different from either sampled-data control or event-based control. Because in both sampled-data and event-based control, the controller $u(t)$ in any instant span $[t_i, t_{i+1})$ is defined by the data (e.g., $x(t_i)$) on the sample time t_i , i.e., $u(t) = u(t_i), \forall t \in [t_i, t_{i+1})$, so the controller is a constant in time span $[t_i, t_{i+1})$; while in this note, the controller $u(t)$ is defined by the data (e.g., $x_j(t) - \pi(t)$) at time t ,

therefore, the controller is time-varying in any time span $[t_i, s_i]$, and will vanish in time span (s_i, t_{i+1}) , $i = 0, 1, 2, \dots$.

B. Synchronization With Static Intermittent Control

In this subsection, we will find sufficient conditions for network (2) reaching synchronization.

Firstly, by [37], $\hat{A}$ is diagonally stable, i.e., there exists a positive definite diagonal matrix

$$\Phi = \text{diag}(\phi_1, \dots, \phi_N) \quad (5)$$

such that $\Phi \hat{A} + \hat{A}^T \Phi < 0$. In the following, without loss of generality, we always assume that $\max_{j=1}^N \phi_j = 1$.

Remark 3: To determine Φ , one can use Matlab LMI toolbox; or choose $[\phi_1, \dots, \phi_N]^T$ the normalized left eigenvector of eigenvalue 0 for matrix A (see [10]).

Theorem 1: Suppose all the rest widths $t_{i+1} - s_i$, $i = 0, 1, 2, \dots$, be bounded, function $f(\cdot) \in \text{QUAD}(P, \Delta)$, and coupling matrix $A \in A1$, denote

$$\bar{A} = (\Phi \hat{A})^s + a_1 \Phi, \quad \underline{A} = (\Phi A)^s + a_1 \Phi \quad (6)$$

where $a_1 = \max_{m=1}^n \delta_m / \gamma_m$, $a_2 = 2 \min_{m=1}^n \gamma_m \lambda_N(\bar{A})$, and $a_3 = 2 \max_{m=1}^n \gamma_m \lambda_N(\underline{A})$. Then the coupled network (2) with aperiodically intermittent pinning control can realize synchronization if

$$\lim_{l \rightarrow +\infty} \left[a_2 \sum_{i=0}^l (s_i - t_i) + a_3 \sum_{i=0}^l (t_{i+1} - s_i) \right] = -\infty. \quad (7)$$

In particular, when $f(\cdot) = 0$, it can realize consensus.

The proof will be given in Appendix A.

In case that the intermittent pinning scheme is periodic, i.e., for any $i = 0, 1, \dots$, $s_i - t_i \equiv \theta$, $t_{i+1} - t_i \equiv \omega$, where θ and ω are positive scalars. Then, under $A \in A1$, network (2) can realize synchronization if $a_2\theta + a_3(\omega - \theta) < 0$, where a_2 and a_3 are defined in the Theorem 1. More detailed discussions can be found in [38].

Remark 4: From condition (19), it can be seen that the convergence rate is determined by

$$\left[a_2 \sum_{i=0}^l (s_i - t_i) + a_3 \sum_{i=0}^l (t_{i+1} - s_i) \right] / [(l+1)\omega].$$

In case a_2 and a_3 are fixed, the larger of the existence percentage of intermittent control $\sum_{i=0}^l (s_i - t_i) / [(l+1)\omega]$, the better the convergence rate is. On the other hand, when the parameters s_i, t_i are fixed, by the definitions of $\bar{A}$ and $\underline{A}$, we can conclude that a_2 must be negative, i.e., $\lambda_N(\bar{A}) < 0$. Thus, matrix $\bar{A}$ should be negative definite. Combining with the fact that $(\Phi \hat{A})^s$ is negative definite, one way to achieve synchronization is to make matrix $(\Phi \hat{A})^s$ large enough. Details are presented and rigorously proved in the Section III-C.

C. Synchronization With Adaptive Intermittent Control

In this subsection, we introduce the adaptive method to intermittent control gain, and discuss the corresponding synchronization problem. Before giving the theoretical results, we present and prove a useful lemma at first.

Lemma 2: Suppose

$$\dot{x}(t) = M(x(t)) - H(t)x(t) \quad (8)$$

where $x(t) \in R$, $M(\cdot) : R \rightarrow R$ is continuous and satisfies $xM(x) \leq L_M x^2$, where $L_M > 0$, and $M(0) = 0$. Function $H(t) : \{0 \cup$

$R_+\} \rightarrow R$ is the adaptive intermittent feedback control gain defined as follows:

$$H(t) = \begin{cases} 0 & t = 0 \\ H(s_i) & t = t_{i+1}, \quad i = 0, 1, 2, \dots \\ 0 & t \in (s_i, t_{i+1}), \quad i = 0, 1, 2, \dots \end{cases} \quad (9)$$

and

$$\dot{H}(t) = x^2(t), \quad t \in [t_i, s_i], \quad i = 0, 1, 2, \dots \quad (10)$$

If there exist two positive scalars $0 < \theta < \omega$, such that

$$\begin{cases} \inf_i (s_i - t_i) & = \theta > 0 \\ \sup_i (t_{i+1} - t_i) & = \omega < +\infty \end{cases} \quad (11)$$

$i = 0, 1, 2, \dots$. Then, $\lim_{t \rightarrow \infty} x(t) = 0$.

The proof will be given in Appendix B.

Now, let us consider N linearly coupled networks with an adaptive aperiodically intermittent pinning control, whose dynamical behaviors can be described as follows:

$$\begin{cases} \dot{x}_1(t) = f(x_1(t)) + H(t) \sum_{k=1}^N a_{1k} \Gamma x_k(t) \\ \quad + H(t) \cdot c \cdot \Gamma (\pi(t) - x_1(t)) \\ \dot{x}_j(t) = f(x_j(t)) + H(t) \sum_{k=1}^N a_{jk} \Gamma x_k(t) \\ \quad j = 2, 3, \dots, N \\ \quad \text{if } t \in [t_i, s_i], \quad i = 0, 1, 2, \dots \\ \dot{x}_j(t) = f(x_j(t)) + \sum_{k=1}^N a_{jk} \Gamma x_k(t) \\ \quad j = 1, 2, \dots, N \\ \quad \text{if } t \in (s_i, t_{i+1}), \quad i = 0, 1, 2, \dots \end{cases} \quad (12)$$

With the same notations as those in Theorem 1, we have

$$\begin{cases} \dot{E}(t) = F(E(t)) + H(t)(\hat{A} \otimes \Gamma)E(t) \\ \quad \text{if } t \in [t_i, s_i], \quad i = 0, 1, 2, \dots \\ \dot{E}(t) = F(E(t)) + (A \otimes \Gamma)E(t) \\ \quad \text{if } t \in (s_i, t_{i+1}), \quad i = 0, 1, 2, \dots \end{cases} \quad (13)$$

where the adaptive control gain defined in (9) with the adaptive rule: for $t \in [t_i, s_i]$, $i = 0, 1, \dots$

$$\dot{H}(t) = h \cdot \sum_{j=1}^N \|x_j(t) - \pi(t)\|^2; \quad h > 0. \quad (14)$$

Applying Lemma 2 to (12), we can state the following theorem for linearly coupled networks concerning the synchronization with adaptive intermittent control.

Theorem 2: Suppose function $f(\cdot)$ satisfies the QUAD condition, if the condition (11) holds, then coupled networks (12) with adaptive rules (9), (14) can realize synchronization.

The proof will be given in Appendix C.

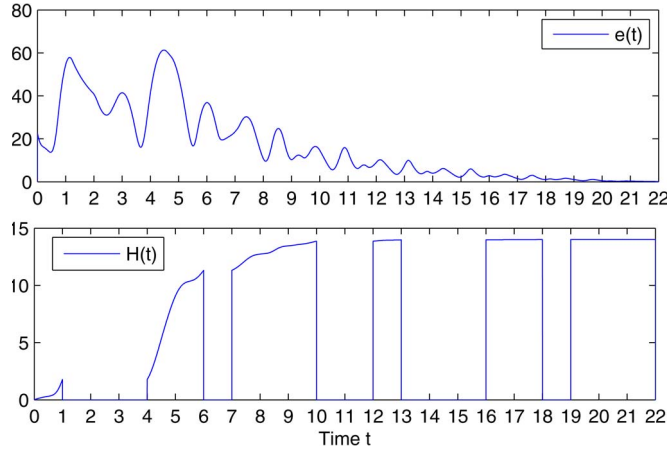


Fig. 2. Top: the trajectories of error $e(t)$, which means that synchronization is realized. Bottom: the trajectory of adaptive intermittent control gain $H(t)$ with $h = 0.0023$.

IV. NUMERICAL EXAMPLES

In this section, we give some numerical simulations to demonstrate the effectiveness of the theoretical results.

Assume the target trajectory $\pi(t)$ is described by a 3-D cellular neural network

$$\dot{\pi}(t) = f(\pi) = -\pi(t) + \begin{pmatrix} -1.25 & -3.2 & -3.2 \\ -3.2 & 1.1 & -4.4 \\ -3.2 & 4.4 & 1 \end{pmatrix} L(\pi(t))$$

where $\pi(t) = (\pi^1(t), \pi^2(t), \pi^3(t))^T$, $L(\pi(t)) = (l(\pi^1(t)), l(\pi^2(t)), l(\pi^3(t)))^T$, $l(v) = (|v+1| - |v-1|)/2$. Numerical simulations show that this neural network has a double-scrolling chaotic attractor [40].

The coupled network is a small-world network composed of 500 oscillators, the connection weights are chosen randomly in $[0, 1]$ with uniform distribution.

In the aperiodically intermittent control strategy, let $s_i - t_i \geq 1$, $t_{i+1} - t_i \leq 6$, $i = 0, 1, 2, \dots$.

Suppose the small-world network is linearly coupled, the intermittent controller with control gain is only added to node 1, then the network can be described by (12).

As reported in [8], function $f(\cdot)$ satisfies the QUAD condition. Moreover, let $\Gamma = I$.

Apply the adaptive approach to the intermittent control, where the adaptive control gain and the adaptive rule are described by (9) and (14), respectively.

In the simulation, $e(t)$ is used to denote the synchronization errors in the network as follows:

$$e(t) = \sqrt{\sum_{j=1}^{500} \|x_j(t) - \pi(t)\|^2 / 500}. \quad (15)$$

The fourth-order Runge-Kutta scheme is used to solve all the ordinary differential equations in our numerical simulations with step-size 0.001. Simulation shows that synchronization can be finally realized, i.e., the error $e(t) \rightarrow 0$, and $\dot{H}(t) \rightarrow 0$, as $t \rightarrow +\infty$. See Fig. 2.

V. CONCLUSION

In this note, synchronization problem under aperiodically intermittent control is investigated. We first give a description of aperiodically intermittent control strategy, which is the main difference of this

note with previous works. Then, using the Lyapunov function, we present some sufficient synchronization criteria for linearly coupled networks under aperiodically intermittent control with static control gain. Moreover, as for aperiodically intermittent control gain, we propose a general adaptive rule and rigorously prove the synchronization problem under this adaptive scheme. Simulations are finally given to demonstrate the effectiveness of the proposed scheme.

APPENDIX A PROOF OF THEOREM 1

Proof: At first, from the definition of positive diagonal matrices Δ and Γ , one can get, for any $m = 1, \dots, n$, $\delta_m = \gamma_m \cdot \delta_m / \gamma_m$, therefore, $\Delta \leq a_1 \Gamma$, where $a_1 = \max_{m=1}^n \delta_m / \gamma_m$.

Define the Lyapunov function as

$$V(t) = \frac{1}{2} E(t)^T (\Phi \otimes P) E(t) \quad (16)$$

where P is defined in Definition 3. Then its derivative with respect to time t along with solutions of (3) can be calculated as follows.

When $t \in [t_i, s_i]$, $i = 0, 1, 2, \dots$

$$\begin{aligned} \dot{V}(t) &= E(t)^T (\Phi \otimes P) \dot{E}(t) \\ &= E(t)^T (\Phi \otimes P) (F(E(t)) + (\hat{A} \otimes \Gamma) E(t)) \\ &= \sum_{j=1}^N \phi_j (x_j(t) - \pi(t))^T P (f(x_j(t)) - f(\pi(t))) \\ &\quad + E(t)^T [\Phi \hat{A} \otimes (P\Gamma)] E(t) \\ &\leq E(t)^T [\Phi \otimes (P\Delta)] E(t) + E(t)^T [(\Phi \hat{A})^s \otimes (P\Gamma)] E(t) \\ &\leq E(t)^T [\bar{A} \otimes (P\Gamma)] E(t) \\ &\leq 2 \min_m \gamma_m \lambda_N(\bar{A}) V(t) = a_2 V(t) \end{aligned}$$

therefore

$$V(t) \leq V(t_i) e^{a_2(t-t_i)}. \quad (17)$$

Similarly, when $t \in (s_i, t_{i+1})$, with the similar analysis, one has

$$\dot{V}(t) \leq E(t)^T [\underline{A} \otimes (P\Gamma)] E(t) \leq a_3 V(t)$$

therefore,

$$V(t) \leq V(s_i) e^{a_3(t-s_i)}. \quad (18)$$

Combining with inequalities (17) and (18), by induction

$$\begin{aligned} V(t_{l+1}) &\leq V(s_l) e^{a_3(t_{l+1}-s_l)} \\ &\leq V(t_l) e^{a_2(s_l-t_l) + a_3(t_{l+1}-s_l)} \\ &\leq V(t_{l-1}) \exp \{ [a_2(s_{l-1}-t_{l-1}) + a_3(t_l-s_{l-1})] \\ &\quad + [a_2(s_l-t_l) + a_3(t_{l+1}-s_l)] \} \leq \dots \\ &\leq V(0) e^{a_2 \sum_{i=0}^l (s_i-t_i) + a_3 \sum_{i=0}^l (t_{i+1}-s_i)}. \end{aligned} \quad (19)$$

From condition (7) and inequality (19), one can get that: for time series $\{V(t_l), l = 0, 1, 2, \dots\}$, $\lim_{l \rightarrow +\infty} V(t_l) = 0$. Recalling inequalities (17) and (18), we have: as time $t \rightarrow +\infty$, $V(t) \rightarrow 0$. So, $e_j(t) \rightarrow 0$, $j = 1, \dots, N$, i.e., synchronization is realized. The proof is completed. ■

APPENDIX B PROOF OF LEMMA 2

Proof: Clearly, the time series $\{H(t_i), i = 0, 1, 2, \dots\}$ is positive and monotone increasing.

Denote $W(t) = x^2(t)$, then

$$\dot{W}(t) = 2[x(t)M(x(t)) - H(t)x^2(t)]. \quad (20)$$

Case 1. There exists some $i^* > 0$, such that $H(t_{i^*}) \geq H^*$, where $H^* > 0$ is a scalar satisfying

$$\lim_{i \rightarrow +\infty} \left[(L_M - H^*) \sum_{k=0}^i (s_k - t_k) + L_M \sum_{k=0}^i (t_{k+1} - s_k) \right] = -\infty. \quad (21)$$

Here, from (11), one can easily get that such H^* does exist.

Then, for $t \in [t_i, s_i]$ with $i > i^*$, we have

$$\dot{W}(t) \leq 2(L_M - H^*)W(t)$$

so

$$W(t) \leq W(t_i)e^{2(L_M - H^*)(t - t_i)}. \quad (22)$$

Instead, when $t \in (s_i, t_{i+1})$, we have

$$\dot{W}(t) \leq 2L_M W(t) \quad (23)$$

so

$$\begin{aligned} W(t) &\leq W(s_i)e^{2L_M(t - s_i)} \\ &\leq W(t_i)e^{2[(L_M - H^*)(s_i - t_i) + L_M(t - s_i)]}. \end{aligned} \quad (24)$$

Therefore, by induction

$$W(t_{i+1}) \leq W(t_i)e^{2[(L_M - H^*)(s_i - t_i) + L_M(t - s_i)]} \leq \dots$$

$$\begin{aligned} &\leq W(t_{i^*}) \exp \left\{ 2 \left[(L_M - H^*) \sum_{k=i^*}^i (s_k - t_k) \right. \right. \\ &\quad \left. \left. + L_M \sum_{k=i^*}^i (t_{k+1} - s_k) \right] \right\}. \end{aligned}$$

From the condition (21), we can get that: for time series $\{W(t_i), i = 0, 1, \dots\}$

$$\lim_{i \rightarrow +\infty} W(t_i) = 0.$$

Combining (22) and (24), we conclude that for all $t \in [t_i, t_{i+1}]$, $i = 0, 1, \dots$, $\lim_{t \rightarrow +\infty} W(t) = 0$, which means $\lim_{t \rightarrow +\infty} x(t) = 0$.

Case 2. For all $t > 0$, $H(t) < H^*$. In this case, we have

$$\sum_{i=0}^{\infty} \int_{t_i}^{s_i} x^2(r) dr < \infty$$

therefore

$$\int_{t_i}^{s_i} x^2(r) dr \rightarrow 0, \text{ as } i \rightarrow +\infty. \quad (25)$$

We claim $\lim_{t \rightarrow \infty} x(t) = 0$. Otherwise, for any constant $\epsilon > 0$, there exist a sequence $T_1 < \dots < T_n \dots$, such that

$T_k < [t_{i-1}, t_{i+1}] < T_{k+1}$, and a sequence $t'_k \in [t_i, t_{i+1}]$, such that $x^2(t'_k) \geq \epsilon$.

From (23), we have $\dot{W}(t) \leq 2L_M W(t)$, which implies that for any $t \leq t'_k$

$$x^2(t'_k) \leq x^2(t)e^{2L_M(t'_k - t)} \quad (26)$$

i.e.,

$$x^2(t) \geq x^2(t'_k)e^{-2L_M(t'_k - t)} \geq \epsilon e^{-2L_M(t'_k - t)}. \quad (27)$$

Therefore, in the interval $t \in [t_{i-1}, t_i]$, we have $x^2(t) \geq \epsilon e^{-4L_M \omega}$, which implies

$$\int_{t_{i-1}}^{s_{i-1}} x^2(r) dr \geq \epsilon \theta e^{-4L_M \omega} \quad (28)$$

a contradiction to (25). Therefore, in this case, we still have $\lim_{t \rightarrow \infty} x(t) = 0$. The proof is completed. ■

APPENDIX C PROOF OF THEOREM 2

Proof: At first, define a function

$$\bar{F}(E(t)) = \begin{cases} F(E(t)), & t \in [t_i, s_i]; \\ F(E(t)) + (A \otimes \Gamma)E(t), & t \in (s_i, t_{i+1}) \end{cases}$$

and define $\bar{L} = \max(2a_1, a_3)$, where a_1 and a_3 are defined in Theorem 1. Then, from the analysis in Theorem 1, we have

$$E(t)^T(\Phi \otimes P)\bar{F}(E(t)) \leq \bar{L}E(t)^T(\Phi \otimes P)E(t). \quad (29)$$

Now, (13) can be stated as

$$\dot{E}(t) = \bar{F}(E(t)) + H(t)(\hat{A} \otimes \Gamma)E(t). \quad (30)$$

Define $\bar{W}(t) = E(t)^T(\Phi \otimes P)E(t)$, where $\Phi = \text{diag}(\phi_1, \dots, \phi_N)$ is defined in (5), then

$$\begin{aligned} \dot{\bar{W}}(t) &= 2E(t)^T(\Phi \otimes P)(\bar{F}(E(t)) + H(t)(\hat{A} \otimes \Gamma)E(t)) \\ &= 2E(t)^T(\Phi \otimes P)\bar{F}(E(t)) \\ &\quad + 2H(t)E(t)^T(\Phi \otimes P)(\hat{A} \otimes \Gamma)E(t) \\ &\leq 2\bar{W}(t) - 2|\lambda_N((\Phi \hat{A})^s)| \cdot \min_{1 \leq m \leq n} \gamma_m \cdot H(t) \cdot \bar{W}(t). \end{aligned}$$

Repeating the proof of Lemma 2, we can obtain $\lim_{t \rightarrow \infty} \bar{W}(t) = 0$, which means $\lim_{t \rightarrow \infty} E(t) = 0$, i.e., synchronization is finally realized. ■

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